

School of Natural Sciences and Mathematics

Molecular Biology and Healthcare Management (Double Major) (BS)

Bachelor of Science in Molecular Biology and Healthcare Management (Double Major)

[Degree Requirements](#) (148 semester credit hours)¹

JSOM Faculty

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Professors: Ashiq Ali, Gary Bolton, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Kyle Hyndman, Varghese S. Jacob, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Zhiang (John) Lin, Sumit K. Majumdar, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Mike W. Peng, Hasan Pirkul, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

Associate Professors: Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Livia Markóczy, Ramachandran (Ram) Natarajan, Naim Bugra Ozel, Young U. Ryu, Harpreet Singh, Upender Subramanian, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

Assistant Professors: Sheen Levine, Radha Mookerjee, Alejandro Rivera Mesias, Guihua Wang, Steven Xiao

Associate Professors Emeriti: J. Richard Harrison, Jane Salk, David J. Springate

Clinical Professors: John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Howard Dover, Randall S. Guttery, William Hefley, Marilyn Kaplan, Peter Lewin, Jeffrey Manzi, John McCracken, Daniel Rajaratnam, Kannan Ramanathan, Mark Thouin, Habte Woldu, Fang Wu

Clinical Associate Professors: Shawn Alborz, Dawn Owens, Carolyn Reichert

Clinical Assistant Professors: Moran Blueshtein, Jeffery (Jeff) Hicks, Liping Ma, Ravi Narayan, Parneet Pahwa, Nassim Sohaee

Professors of Instruction: Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Matt Polze, Luell (Lou) Thompson

Associate Professors of Instruction: Judd Bradbury, Amal El-Ashmawi, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Hubert Zydorek

Assistant Professors of Instruction: Daniel Karnuta, Joseph Mauriello

Professors of Practice: Tiffany A. Bortz, Alexander Edsel, Rajiv Shah, Keith Thurgood

Associate Professors of Practice: Richard Bowen, Jackie Kimzey, Margaret Smallwood, Steven Solcher, Kathy Zolton

Assistant Professors of Practice: Edward Meda, Timothy Stephens, Robert Wright

NSM Faculty

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Professors: Juan E. González, Tae Hoon Kim, Kelli Palmer, Lawrence J. Reitzer, Donal Skinner, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Joseph Boll, Nicole De Nisco, Nikki Delk, Tian Hong, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Brandon Kim, Erica Sanchez, Darshan Sapkota, Yuki Shindo, Jon Sin

Professors Emeriti: Hans Bremer, Lee A. Bulla, Rockford Draper, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison, Eberhard Voit

Professors of Instruction: Mehmet Candas, Elizabeth Pickett, Uma Srikanth

Associate Professors of Instruction: Meenakshi Maitra, Jing Pan, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu, Wen-Ho Yu

Assistant Professors of Instruction: Stephanie Boyd, Andy Cheshire, Anne Davenport, Matthew Esparza, Amy Jo Gomez, Yi Huang, Li Liu, Simbarashe Mazambani, Iti Mehta, Ritu Mishra, Ramesh Padmanabhan, Narges Salamat

Research Associate Professor: Ru-Hung Wang

Lecturer: Kathleen McRoy

Students must meet all course prerequisites.

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I ^{2, 3, 4, 5, 6}

or [MATH 2413](#) Differential Calculus ^{2, 3, 4, 5, 6}

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I ³

or [CHEM 1315](#) Honors Freshman Chemistry I ³

[CHEM 1312](#) General Chemistry II ³

or [CHEM 1316](#) Honors Freshman Chemistry II ³

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses

Social and Behavioral Sciences: 3 semester credit hours

Choose one of the following:⁷ _{_}

[BA 1310](#) Making Choices in Free Market Systems^{2, 3} _{_}

[BA 1320](#) Business in a Global World^{2, 3} _{_}

[ECON 2301](#) Principles of Macroeconomics^{2, 3} _{_}

[ECON 2302](#) Principles of Microeconomics^{2, 3} _{_}

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses

Component Area Option: 6 semester credit hours

Choose two of the following:

[MATH 2419](#) Calculus II^{3, 4, 8} _{_ _}

or [MATH 2414](#) Integral Calculus^{3, 4, 8} _{_ _}

[BA 1310](#) Making Choices in Free Market Systems^{2, 3} _{_}

[BA 1320](#) Business in a Global World^{2, 3} _{_}

[ECON 2301](#) Principles of Macroeconomics^{2, 3} _{_}

[ECON 2302](#) Principles of Microeconomics^{2, 3} _{_}

Or select any 6 semester credit hours from [Component Area Option Core](#) courses

II. Major Requirements: 88 semester credit hours

Biology Major Preparatory Courses: 20 semester credit hours beyond Core Curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I³ _{_}

or [CHEM 1315](#) Honors Freshman Chemistry I³ _{_}

[CHEM 1312](#) General Chemistry II₃

or [CHEM 1316](#) Honors Freshman Chemistry II₃

[CHEM 2323](#) Introductory Organic Chemistry I₂

or [CHEM 2327](#) Honors Organic Chemistry I₂

[CHEM 2325](#) Introductory Organic Chemistry II₂

or [CHEM 2328](#) Honors Organic Chemistry II₂

[CHEM 2233](#) Introductory Organic Chemistry Laboratory₂

or [CHEM 2237](#) Honors Organic Chemistry Laboratory₂

[MATH 2417](#) Calculus I_{2, 3, 4, 5, 6}

or [MATH 2413](#) Differential Calculus_{2, 3, 4, 5, 6}

[MATH 2419](#) Calculus II_{3, 4, 8}

or [MATH 2414](#) Integral Calculus_{3, 4, 8}

[PHYS 2325](#) Mechanics

and [PHYS 2125](#) Physics Laboratory I

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat₉

[PHYS 2326](#) Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

Biology Core Courses: 28 semester credit hours

[BIOL 2111](#) Introduction to Modern Biology Workshop I₂

[BIOL 2112](#) Introduction to Modern Biology Workshop II₂

[BIOL 2281](#) Introductory Biology Laboratory₂

or BIOL2v81 Course-based Undergraduate Research Experience in Biology₂

[BIOL 2311](#) Introduction to Modern Biology I₂

[BIOL 2312](#) Introduction to Modern Biology II₂

[BIOL 3401](#) Genetics

[BIOL 3402](#) Molecular and Cell Biology

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

[BIOL 3462](#) Biochemistry II

or [CHEM 3462](#) Biochemistry II

or [BIOL 3335](#) Microbial Physiology

[BIOL 3380](#) Biochemistry Laboratory

[BIOL 4461](#) Biophysical Chemistry

Business Major Preparatory Courses: 12 semester credit hours beyond Core Curriculum

[ACCT 2301](#) Introductory Financial Accounting₂

[ACCT 2302](#) Introductory Management Accounting₂

[BLAW 2301](#) Business and Public Law₂

[OPRE 3360](#) Managerial Methods in Decision Making Under Uncertainty₁₀

or [STAT 2332](#) Introductory Statistics for Life Sciences

Choose two of the following if not completed in Core Curriculum:₇:

[BA 1310](#) Making Choices in Free Market Systems_{2, 3}

or [BA 1320](#) Business in a Global World_{2, 3}

or [ECON 2301](#) Principles of Macroeconomics_{2, 3}

or [ECON 2302](#) Principles of Microeconomics_{2, 3}

Business Core Courses: 28 semester credit hours

[BA 1105](#) Professional Development (JSOM first-time-in-college freshmen are required to take [BA 1105](#) in their first semester)

or [BA 3105](#) Professional Development (Transfer students and students new to JSOM are required to take [BA 3105](#) in their first semester)

[BCOM 3300](#) Business Communication

[BUAN 3301](#) AI in Business

[FIN 3320](#) Business Finance

[IMS 3310](#) International Business

[ITSS 3300](#) Information Technology for Business

[MKT 3300](#) Principles of Marketing

[OBHR 3310](#) Organizational Behavior

or [OBHR 3330](#) Human Resource Management in Business

[OPRE 3310](#) Supply Chain and Operations Management

Experiential Learning

A capstone course is required for all undergraduate JSOM students.

[BPS 4395](#) Capstone Senior Project - Business

A practicum experience of at least 160 working hours is required, with registration in the course below.

[HMGT 4090](#) Healthcare Management Internship¹¹

Students are required to complete 100 hours of community engagement experience with registration in the course below.

[BA 4095](#) Social Sector Engagement and Community Outreach Practicum¹²

III. Elective Requirements: 18 semester credit hours

Guided Electives: 18 semester credit hours

Healthcare Management Core Courses: 15 semester credit hours

[HMGT 3301](#) Healthcare Management

[HMG 3310](#) Healthcare Regulation and Economics

[HMG 3311](#) Healthcare Financial Analysis

[HMG 3373](#) Healthcare Database Management

[HMG 4321](#) Healthcare Information Systems

Biology (3 semester credit hours):

[BIOL 4380](#) Cell and Molecular Biology Laboratory¹³

or [BIOL 3V96](#) Undergraduate Research in Molecular and Cell Biology

or [BIOL 4391](#) Senior Research in Molecular and Cell Biology

or [BIOL 4399](#) Senior Honors Research for Thesis in Molecular and Cell Biology

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Degree is 148-151 semester credit hours if students are required to take NATS 1101.
2. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
3. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
4. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option Core, and 2 semester credit hours of Calculus are counted as Biology Major Preparatory Courses.
5. MATH 2413 is required for the Healthcare Management/Molecular Biology prescribed double major. Students may elect to substitute MATH 2417 for MATH 2413.
6. To make timely degree progress, students should complete MATH 2413 or MATH 2417 by the end of their first semester at UT Dallas. Students who will not meet this requirement should contact their academic advisor to discuss their degree timeline.
7. Certain courses listed are prerequisites for major core, major concentration, or major related courses. Choose accordingly.
8. Students may elect to substitute MATH 2419 for MATH 2414.
9. Students who complete PHYS 2421 do not need to complete PHYS 2125.
10. Students may use CS 3341, ENGR 3341, SE 3341, STAT 3341, STAT 2332, STAT 3355, STAT 3360, or STAT 4351 as a substitute for OPRE 3360.

11. Students may fulfill the internship requirement with BA 4090, HMGT 4090, or HMGT 4V90 (1-3 semester credit hours). The zero semester credit hour courses HMGT 4090 or BA 4090 are recommended as the most efficient way to satisfy this requirement. If the variable credit internship option is taken, semester credit hours apply as free elective credit towards degree requirements.

12. Students may fulfill the community engagement requirement with BPS 4396 or BA 4095, IMS 4335, ENTP 4340, MKT 4360, or OPRE 4391. The zero semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this requirement. If the variable credit community engagement option is taken, semester credit hours apply as free elective credit towards degree requirements.

13. Requires permission of the Biology Undergraduate Advisor to ensure training in recombinant DNA analysis.

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